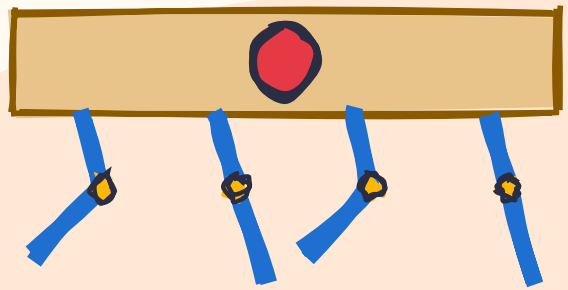
Young Innovators · Class 26

# Walking Linkage Robot

## No wheels. No code. Just clever sticks and pins.

Four-bar | Foot path | Phased legs



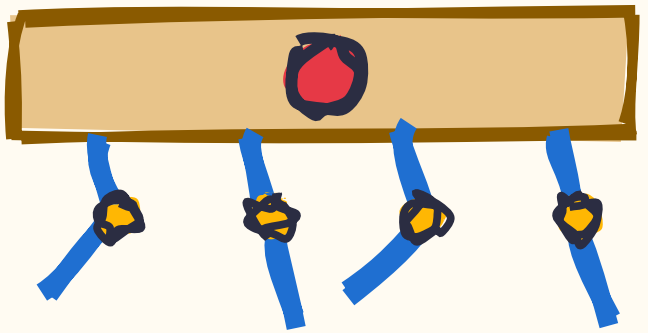
Did you know?
Theo Jansen's Strandbeest creatures walk along Dutch beaches powered only by wind, using one clever linkage repeated many times.

Let's think...

# Why don't robots just use wheels?

**Wheels vs legs**

Wheels are simple and fast on roads. But on sand, stairs, rocks and mud, wheels get stuck while animals keep walking. Can we make legs that walk using just ONE turning crank?



**Wheels are best!**

**Too complicated!**

**Linkages!**

**Today:** you build a **walking robot** from card, skewers and split pins: one crank, many legs.

Learning objectives

# By the end of this class I can...

| | | |
|---|---|---|
| 1 | **Links and pivots:** name the parts of a **four-bar linkage** (crank, coupler, rocker, ground) and how it changes motion | **Proof:** my card four-bar turns a crank into a rocking motion |
| 2 | **Foot path:** trace the **path** of a point on a linkage and explain why a walking foot needs a flat bottom | **Proof:** I drew my leg's foot path |
| 3 | **Walker:** build a **multi-leg walker** with legs out of **phase** and measure how far it walks per turn | **Proof:** my walker moves forward when I turn the crank |

Objective 3 · Walker · video

# Video time!




**Theo Jansen's Wind-Powered Sculptures**

The New Yorker · about 2 minutes · youtu.be/Pj-NqWDH2qE

## Watch for...

1. What **powers** the Strandbeest?
2. How do the legs **move** without a motor?
3. Why did Jansen keep **changing** his designs?

Scan to watch

**After the video:** Where else could a walking machine be more useful than wheels?

Objective 1 · Links and pivots

# Links + pivots = a linkage

## The idea

A **link** is a rigid bar. A **pivot** (split pin) lets two links turn. Join four links in a loop and you get a **four-bar linkage**:

**Ground** (grey): fixed frame.

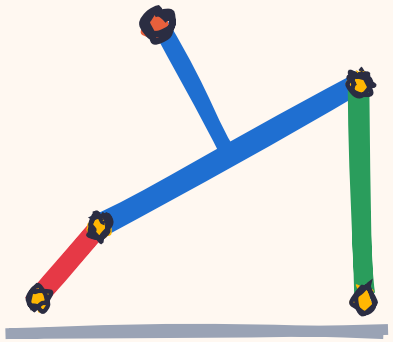
**Crank** (red): turns all the way round.

**Coupler** (blue): the middle link that 'floats'.

**Rocker** (green): swings back and forth.

Round-and-round goes in; rocking comes out.

## Example: a bicycle pedal and your leg



Your thigh, shin, the pedal crank and the bike frame form a four-bar! The pedal goes round; your knee rocks up and down.

Objective 2 · Foot path

# The magic of the coupler path

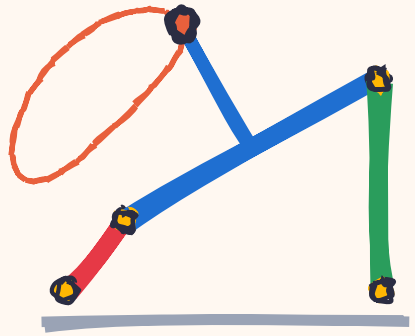
## The idea

Points on the crank draw **circles**. Points on the rocker draw **arcs**. But a point on the **coupler** can draw almost any shape: ovals, teardrops, even D-shapes!

For walking we want a path with a long, **flat bottom** (foot on the ground, pushing back) and a **lifted top** (foot in the air, swinging forward).

Change one link length → change the whole path.

## Example: drawing with a Spirograph



A Spirograph uses gears to draw curly shapes. Linkages are like a Spirograph with sticks: small changes give totally different patterns.

Objective 3 · Walker

# Legs out of phase

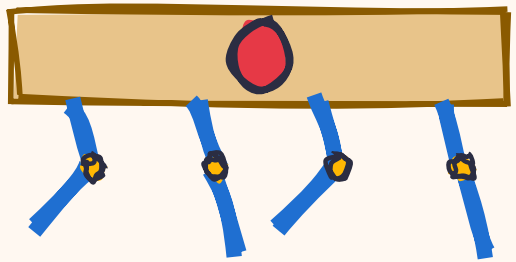
## The idea

One leg alone falls over: its foot spends half the time in the air.

Put legs on the same crank but **180° apart** (out of phase): while leg A is on the ground pushing, leg B is in the air returning. Use **at least 4 legs** (2 each side) so the body is always supported.

Jansen's beasts use 3 legs per crank angle group for a super-smooth walk.

## Example: walking like a human



When your left foot is on the ground pushing, your right foot swings forward. Your legs are 180° out of phase. Try walking with both legs in phase: you hop!

Real world

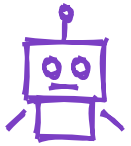
# Linkages everywhere



**Strandbeest**
Wind-powered walking sculptures, Netherlands



**Windscreen wipers**
A four-bar turns motor rotation into sweeping



**Walking robots**
Boston Dynamics legs use linkages + motors



**Bike suspension**
Linkages control how the wheel moves

? Find a linkage in our classroom or at home. What goes in and what comes out?

**Holy numbers**
Jansen used a computer to test thousands of leg lengths. The winning 11 lengths are called his 'holy numbers'.

Quick quiz

# Linkage logic

| Question | Answer |
|---|---|
| 1. Which link goes all the way round? | |
| 2. Which link only swings? | |
| 3. What shape does a crank point draw? | |
| 4. Why legs 180° apart? | |
| 5. What foot-path shape is best for walking? | |

**Answers**

Cover! 1: crank · 2: rocker · 3: a circle · 4: one always on the ground · 5: long flat bottom, lifted top

Part 2 · Make it

# Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human four-bar
2. Our walker kit (per pair)
3. Build a four-bar and trace its path
4. Build one walking leg
5. Make it walk
6. Walker trials
7. Mechanism designer challenges

Class game

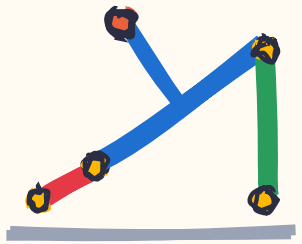
# Human four-bar

## How to play

1. 4 students hold 4 metre rulers by the ends, making a loop (each hand is a pivot).
2. Two students on the 'ground' link stand still.
3. The 'crank' student walks a small circle. Everyone else just follows!
4. A fifth student holds a marker on the coupler and draws on paper taped to the board.

## Then discuss

What shape did the coupler marker draw? What happens if the crank is longer?



Big idea

One input movement, a whole new output movement.

Build it

# Our walker kit (per pair)



**Thick card / foam board**
For links and body



**Split pins (paper fasteners)**
Pivots



**Wooden skewers**
Crank axle



**Leg template**
Printed lengths (class folder)



**Hole punch / bradawl**
Teacher pokes pivot holes



**Gear motor (optional)**
For a motorised walker

Holes at exact link lengths matter more than neat cutting. Measure centre to centre.

**Sharp tools**
Only the teacher uses the bradawl or craft knife.
Skewers: blunt the points with scissors first.

MAKE

Objective 2 · Build step 1 of 3

# Build a four-bar and trace its path

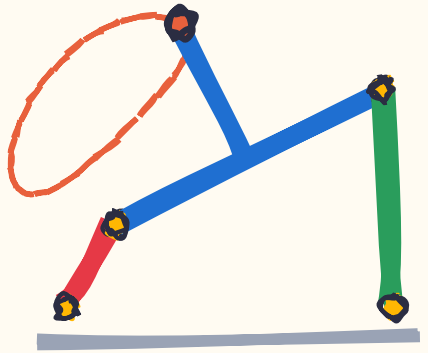
**a** Cut 4 strips: ground 120 mm, crank 35 mm, coupler 110 mm, rocker 80 mm (hole to hole).

**b** Pin them into a loop. Pin the ground strip to a sheet of card on the table.

**c** Turn the crank all the way round. If it jams, check the lengths.

**d** Push a pencil through a hole on a coupler extension. Turn and **draw the path**.

**e** Change the coupler length by 10 mm and draw again. Compare.



**Check:** You have two drawn paths and can describe how they differ.

MAKE

Objective 2 · Build step 2 of 3

# Build one walking leg

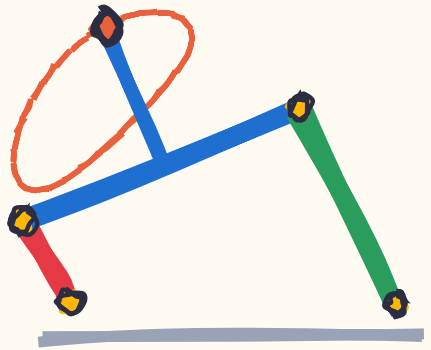
**a** Use the **leg template** (Jansen-style, scaled to card).

**b** Cut every link; mark hole centres with the template.

**c** Teacher punches holes. Assemble with split pins, not too tight (links must swing freely).

**d** Fix the frame points to a card body. Push the skewer crank through.

**e** Turn the crank: does the foot draw a flat-bottomed path?



**Check:** Your single leg's foot moves in a walking path.

Split pin legs can scratch: fold them flat at the back.

MAKE

Objective 3 · Build step 3 of 3

# Make it walk

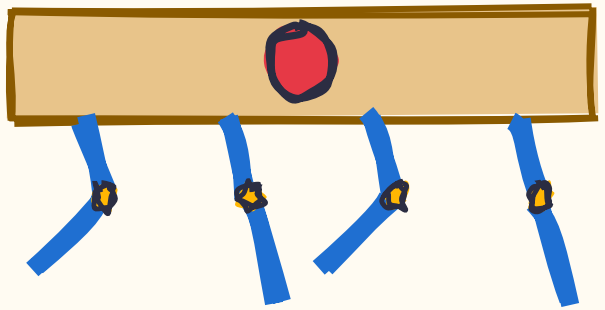
**a** Build 4 identical legs. Mount 2 on each side of the body.

**b** Set crank pins so each side's legs are **180° apart**, and left/right are also opposite.

**c** Turn the skewer crank slowly. Does it step forward?

**d** Measure: how many cm does it travel in 5 crank turns?

**e** Optional: add a gear motor on the crank for a motorised walker.



**Check:** Your walker moves forward when you turn the crank.

Test like an engineer

# Walker trials

| Test | Result | Unit |
|---|---|---|
| Crank turns freely all the way? | | |
| Distance in 5 turns | | cm |
| Distance per turn | | cm |
| Walks on carpet / on paper? | | |
| Change we made + new distance | | cm |

## Debug checklist

Jams halfway? A link is too short or holes are off: re-measure centre to centre.
Slides backwards? Add rubber feet (a bit of balloon or tape).
Wobbles? Pins too loose: add a washer.

Engineers change ONE length at a time and measure again. That's how Jansen found his holy numbers.

Level up!

# Mechanism designer challenges

## Choose your challenge

- **Six legs:** add a third pair. Is it smoother?
- **Wind power:** add a card propeller on the crank. Can wind make it walk?
- **Simulate:** try the Strandbeest linkage simulator online and change lengths.
- **Motor:** add a TT gear motor. How fast can it walk before it trips?

## Think like an engineer

Why do walking machines struggle more than wheels on flat roads, but win on sand?

**Boss level**

Design a linkage that makes a creature's head nod as it walks.

Objective check

# Can I really do it? Prove it.

| | | |
|---|---|---|
| 01 | Point to the crank, coupler, rocker and ground on your four-bar. | ○ ○ ○ |
| 02 | Show your traced foot path. Where is the foot touching the ground? | ○ ○ ○ |
| 03 | Show your walker moving. How far per crank turn? | ○ ○ ○ |

**Rate yourself**

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

**Teacher:** Walker race: 10 crank turns each. Furthest walker wins; smoothest walker gets a design prize.

Show what you know

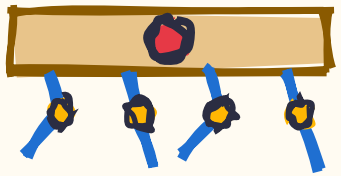
# What did we learn?

- [x] Links + pivots make linkages
- [x] Four-bar: ground, crank, coupler, rocker
- [x] Coupler points draw special paths
- [x] Walking feet need a flat-bottomed path
- [x] Legs out of phase keep the body supported

**Exit ticket:** what does the rocker do while the crank turns all the way round?

At home

Find a linkage at home: scissors lift, folding chair, umbrella, or car wipers.



**Next class:**
Energy Detective: How Much Power Do Gadgets Use?